

**INSTITUT UNIVERSITAIRE DE LA
CÔTE**
School of Engineering and Applied Sciences

**Data Structures Applications
Group Project Assignment**

END OF YEAR PROJECT ASSIGNMENT

Issued: 20 December 2025

Course Information

Course:	CSE523 - Algorithms & Data Structures
Class:	MengBy (Master of Engineering - Batch Y)
Lecturer:	Mr. FOMAZOU TCHINDA
Academic Year:	2025/2026
Weight:	30% of Final Grade
Submission Email:	fomazoumilka@gmail.com

**Department of Computer Science & Engineering
Institut Universitaire de la Côte (IUC)
Douala, Cameroon**

Deadline: 15 January 2026

Contents

Important Notice	2
1 Group Assignments	2
2 Project Descriptions	2
2.1 Project 1: Medical Supply Chain Management	2
2.2 Project 2: Agricultural Market Linkage Platform	3
2.3 Project 3: Mobile Money Fraud Detection	3
2.4 Project 4: Smart Waste Management System	3
2.5 Project 5: Adaptive Learning Platform	3
2.6 Project 6: Smart Public Transit System	3
2.7 Project 7: Livestock Disease Surveillance	3
2.8 Project 8: Cross-Border Trade Platform	4
2.9 Project 9: Smart Grid Management System	4
2.10 Project 10: Tourism Platform	4
3 Technical Requirements	4
3.1 Mandatory Requirements	4
4 Holiday Timeline & Deadlines	4
5 Submission Requirements	5
6 Assessment Criteria	5
7 Academic Integrity	6
Group Registration Form	7
Contact Information	8

Important Notice

URGENT: END OF YEAR ASSIGNMENT

This project constitutes 40% of your final grade for the 2025/2026 academic year. Given the issuance date of 20 December 2025, you have limited time to complete this assignment.

Submit all deliverables to: fomazoumilka@gmail.com

Note: The university will reopen on 6 January 2026. Use the holiday period productively to work on this project.

1 Group Assignments

Below are the assigned projects for each group. Groups consist of 2-3 members as indicated. These assignments are final and cannot be changed.

Table 1: Group Project Assignments (Academic Year 2025/2026)

Group No.	Members	Project Title	Due Date
1	2-3 members	Medical Supply Chain Management System	15 Jan 2026
2	2-3 members	Agricultural Market Linkage Platform	15 Jan 2026
3	2-3 members	Mobile Money Fraud Detection System	15 Jan 2026
4	2-3 members	Smart Waste Management System	15 Jan 2026
5	2-3 members	Adaptive Learning Platform	15 Jan 2026
6	2-3 members	Smart Public Transit System	15 Jan 2026
7	2-3 members	Livestock Disease Surveillance Network	15 Jan 2026
8	2-3 members	Cross-Border Trade Platform	15 Jan 2026
9	2-3 members	Smart Grid Management System	15 Jan 2026
10	2-3 members	Tourism & Cultural Heritage Platform	15 Jan 2026

2 Project Descriptions

2.1 Project 1: Medical Supply Chain Management

Problem: Optimize medical supply distribution across Cameroon's 10 regions.

Data Structures: Graph, Priority Queue, Hash Table, BST, Queue, Stack

Focus: Logistics, emergency prioritization, route optimization

Special Context: Consider rainy season road closures and regional disparities

2.2 Project 2: Agricultural Market Linkage Platform

Problem: Connect farmers with buyers using predictive analytics.

Data Structures: Trie, Graph, Hash Table with Chaining, Segment Tree

Focus: Agriculture, predictive analytics, market matching

Special Context: Focus on major crops: cocoa, coffee, maize, plantains

2.3 Project 3: Mobile Money Fraud Detection

Problem: Real-time fraud detection for mobile money platforms.

Data Structures: Graph, Bloom Filter, Circular Buffer, Trie

Focus: Finance, security, pattern recognition

Special Context: MTN Mobile Money and Orange Money patterns

2.4 Project 4: Smart Waste Management System

Problem: Optimize waste collection routes in Yaoundé/Douala.

Data Structures: Weighted Graph, Priority Queue, Union-Find, Stack

Focus: Urban planning, route optimization, environmental management

Special Context: Account for traffic patterns and market days

2.5 Project 5: Adaptive Learning Platform

Problem: Offline-capable learning platform for rural schools.

Data Structures: Trie, Graph, Hash Table, Stack, Priority Queue

Focus: Education technology, adaptive learning, offline access

Special Context: Consider intermittent electricity and internet access

2.6 Project 6: Smart Public Transit System

Problem: Real-time public transportation tracking and routing.

Data Structures: Graph, Min-Heap, Segment Tree, Bloom Filter

Focus: Transportation, real-time systems, route planning

Special Context: Integrate buses, taxis, and motorcycle taxis (bend-skin)

2.7 Project 7: Livestock Disease Surveillance

Problem: Track and predict livestock disease outbreaks.

Data Structures: Graph, Trie, Segment Tree, Union-Find

Focus: Agriculture, epidemiology, data analysis

Special Context: Focus on cattle, goats, poultry in different regions

2.8 Project 8: Cross-Border Trade Platform

Problem: Facilitate trade between Cameroon and neighboring countries.

Data Structures: Graph, Trie, Segment Tree, Bloom Filter

Focus: E-commerce, international trade, logistics

Special Context: Nigeria, Chad, CAR, Gabon, Equatorial Guinea borders

2.9 Project 9: Smart Grid Management System

Problem: Optimize electricity distribution in Cameroon’s mixed grid.

Data Structures: Graph, Priority Queue, Segment Tree, Union-Find

Focus: Energy management, load balancing, renewable integration

Special Context: Hydro, solar, thermal energy mix; rural micro-grids

2.10 Project 10: Tourism Platform

Problem: Promote tourism with intelligent itinerary planning.

Data Structures: Graph, Trie, Priority Queue, Bloom Filter

Focus: Tourism, recommendation systems, cultural preservation

Special Context: Include Mount Cameroon, Waza Park, cultural festivals

3 Technical Requirements

3.1 Mandatory Requirements

Core Requirements - All Projects
• ✓5+ Data Structures: Implement at least 5 different data structures • ✓Complexity Analysis: Include Big-O analysis for all operations • ✓Cameroonian Context: Address local constraints and realities • ✓Dual Interface: CLI and simple web interface • ✓Testing: Unit tests covering critical paths • ✓Documentation: Technical report (10-15 pages) • ✓Presentation: 5-10 minute video demonstration

4 Holiday Timeline & Deadlines

Table 2: Holiday Period Project Timeline (December 2025 - January 2026)

Date	Deliverable	Notes
20 Dec 2025	Assignment Issued	Today - Project officially assigned
27 Dec 2025	Group Registration	Submit group members and project confirmation
3 Jan 2026	Project Proposal	2-page proposal with design approach (email submission)
10 Jan 2026	Progress Report	Initial implementation with core data structures
15 Jan 2026	Final Submission	Complete project with all deliverables
18-22 Jan 2026	Presentations	Project demonstrations (schedule to be announced)

5 Submission Requirements

Submit the following to fomazoumilka@gmail.com:

Submission Package - Email Requirements

- **Email Subject:** CS405_Group[Number]_[ProjectName]_MengBy2025
- **Example:** CS405_Group3_FraudDetection_MengBy2025
- **Attachments:** Single ZIP file named as above containing:
 - Source code (organized folder structure)
 - Technical documentation (PDF, 10-15 pages)
 - Presentation slides (PDF, 5-10 slides)
 - Demo video (5-10 minutes, MP4 or YouTube link)
 - README file with setup instructions
 - Test cases and sample data
 - Completed Group Registration Form (scanned PDF)

6 Assessment Criteria

Table 3: Grading Breakdown - End of Year 2025/2026

Criterion	Weight	Description
Data Structure Implementation	30%	Correct implementation of 5+ data structures with proper operations
Algorithm Efficiency	25%	Optimal algorithms with proper complexity analysis
System Functionality	20%	Working system that solves the stated problem
Cameroonian Context	15%	Adaptation to local conditions and constraints
Documentation & Presentation	10%	Quality of documentation and video demonstration
Total	100%	

7 Academic Integrity

Academic Integrity Policy - 2025/2026:

- All work must be original and created by your group during this holiday period
- Code plagiarism will result in zero grade and disciplinary action
- Cite all references properly using IEEE format
- Peer assessment forms will be used to evaluate individual contributions
- All submissions will be checked using plagiarism detection software

Group Registration Form

Complete this form, sign, scan, and email to: fomazoumilka@gmail.com
Deadline: 27 December 2025

GROUP REGISTRATION FORM

CS 405 - Data Structures & Algorithms
MengBy Class 2025/2026

Group Information:

- **Assignment Date:** 20 December 2025
- **Group Number:** _____ (from Table 1, page 3)
- **Project Assigned:** _____
- **Final Due Date:** 15 January 2026

Group Members (2-3 members):

1. **Team Leader:** _____ (ID: _____)
Email: _____
2. **Member 2:** _____ (ID: _____)
Email: _____
3. **Member 3:** _____ (ID: _____)
Email: _____

Declaration:

We, the undersigned, confirm that we have read and understood the project requirements for the 2025/2026 academic year. We acknowledge that this project constitutes 40% of our final grade and agree to abide by the academic integrity policy.

Team Leader Signature	Date	Group Number

Member Signatures:

1. -----
2. -----
3. -----

Contact Information - Holiday Period

Communication During Holidays

- **Lecturer:** Mr. FOMAZOU TCHINDA
- **Email:** fomazoumilka@gmail.com
- **Response Time:** Allow 24-48 hours for email responses during holidays
- **Emergency Contact:** Only for technical clarification emergencies
- **Submission Portal:** Email submissions only - no physical submissions
- **University Reopening:** 6 January 2026

Holiday Working Advice:

- Form your groups immediately (by 27 December 2025)
- Use online collaboration tools (GitHub, Google Docs, Zoom)
- Schedule regular virtual meetings with your group
- Test your implementations with sample Cameroonian data
- Start documentation early and update regularly

End of Year 2025/2026 Academic Calendar Note

The university will be closed from 23 December 2025 to 5 January 2026 for the end of year break. Use this time productively to work on your project. All communications will be via email during this period.

Season's Greetings and Productive Holidays!

"The best way to predict the future is to create it."

Happy New Year 2026!
